



Ingesting and Preparing Data

AWS Academy Data Engineering

Introduction

Ingesting and Preparing Data

Module objectives

This module prepares you to do the following:

- Distinguish between the processes of extract, transform, and load (ETL) and extract, load, and transform (ELT).
- Define data wrangling in the context of ingesting data into a pipeline.
- Describe key tasks in each of these data wrangling steps:
 - Discovery
 - Structuring
 - Cleaning
 - Enriching
 - Validating
 - Publishing

Module overview

Presentation sections

- ETL and ELT comparison
- Data wrangling introduction
- Data discovery
- Data structuring
- Data cleaning
- Data enriching
- Data validating
- Data publishing

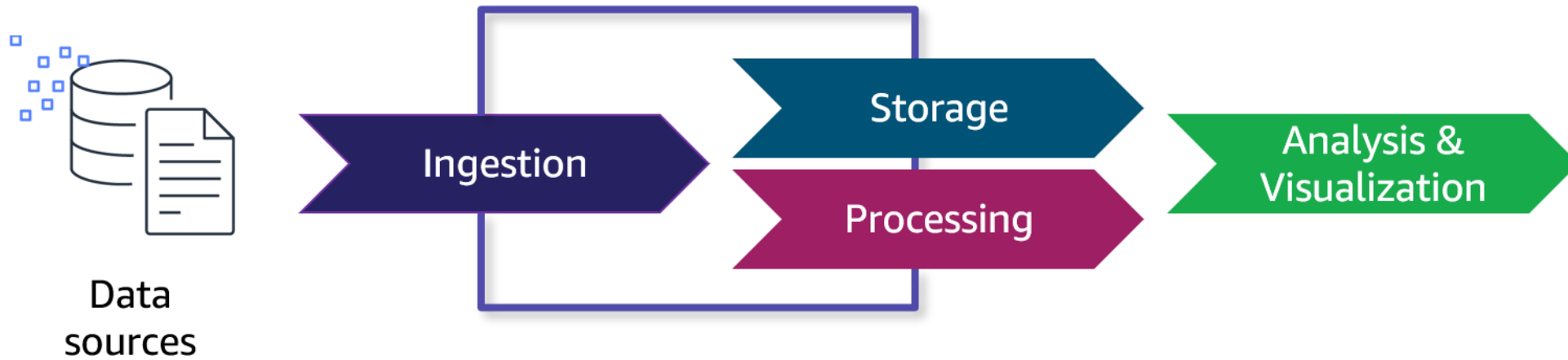
Knowledge checks

- Online knowledge check
- Sample exam question

ETL and ELT comparison

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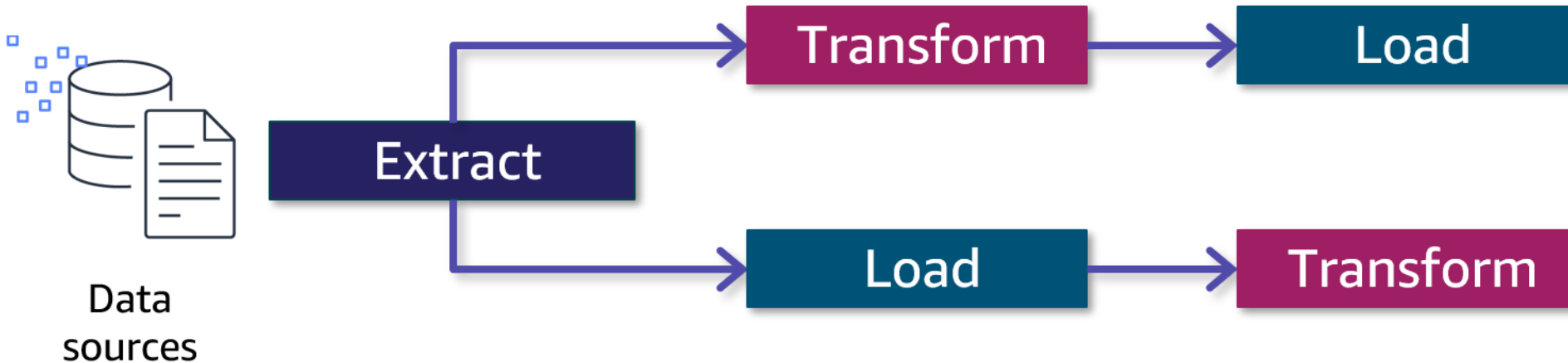
Ingesting data involves storing and processing data



ETL and ELT

Extract transform load (ETL)

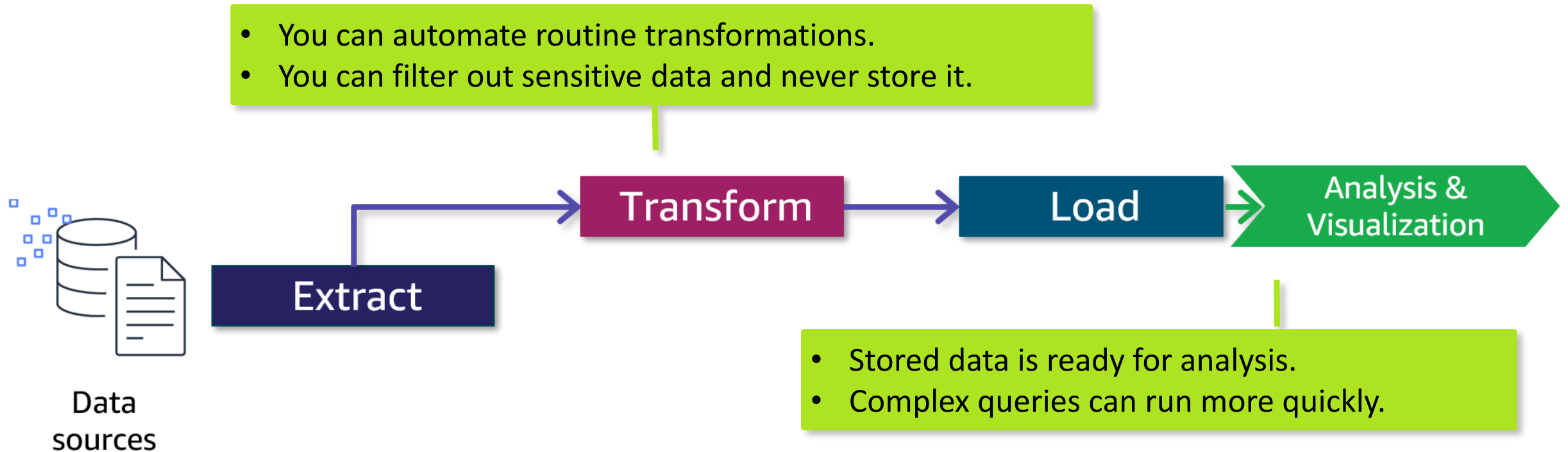
1. **Extract** structured data.
2. **Transform** the data into a format that matches the destination.
3. **Load** the data into structured storage for defined analytics.



Extract load transform (ELT)

1. **Extract** unstructured or structured data.
2. **Load** the data into the data lake in the format that is as close to the raw form as possible.
3. **Transform** the data as needed for analytic scenarios.

Benefits of extract transform load (ETL)

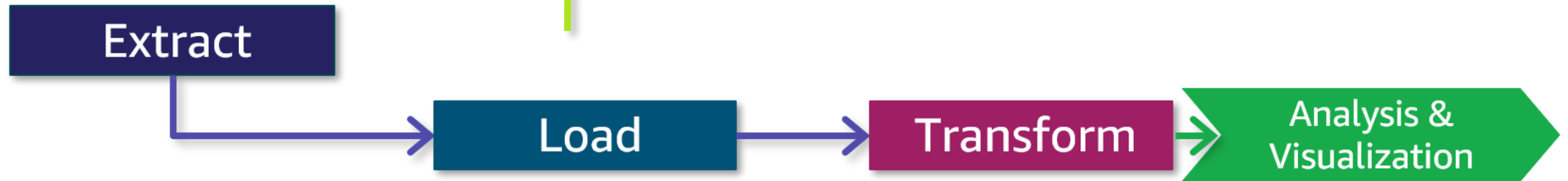


Benefits of extract load transform (ELT)

- The ingestion process can run more quickly.
- You don't invest effort in guessing what people will use.



Data
sources



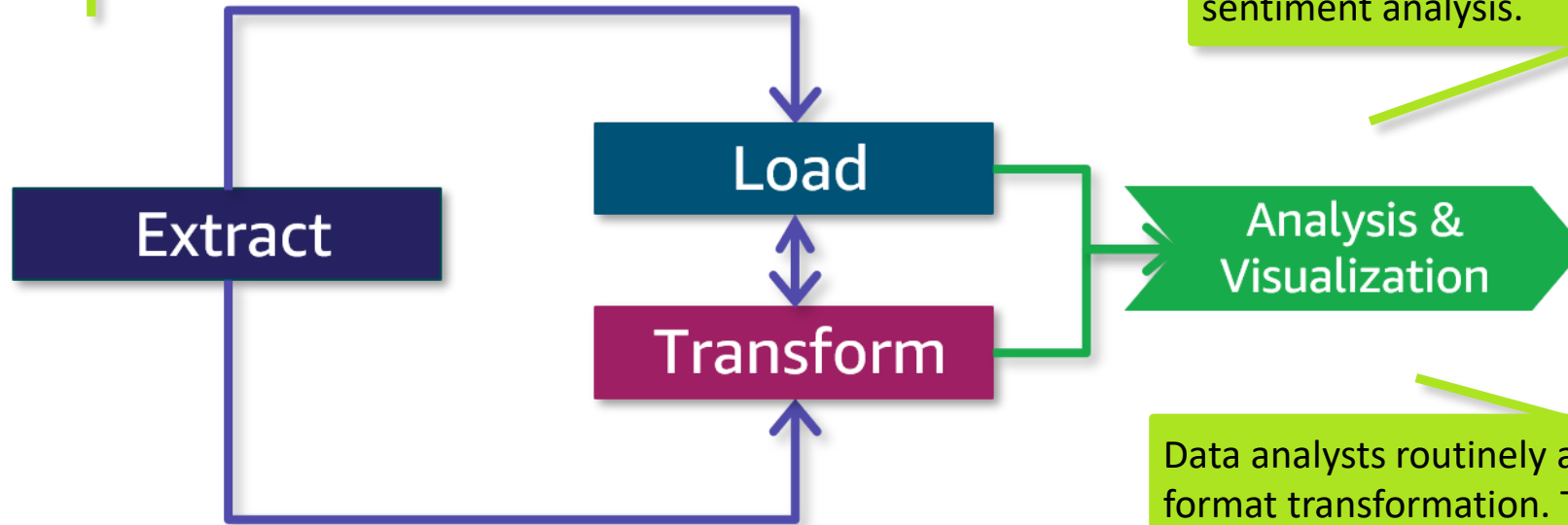
- This method supports one-time, or ad hoc analysis.
- Changes to transformations apply to historical data.

The ingestion process evolves as the pipeline is used

A data engineer automates a nightly batch process to ingest support ticket data into the pipeline.



Data sources



A data scientist requests that comment fields are no longer truncated when support ticket data is ingested. They want all the data for sentiment analysis.

Data analysts routinely apply a common date format transformation. They ask that the transformation be added to ingestion processing.

Key takeaways: ETL and ELT comparison

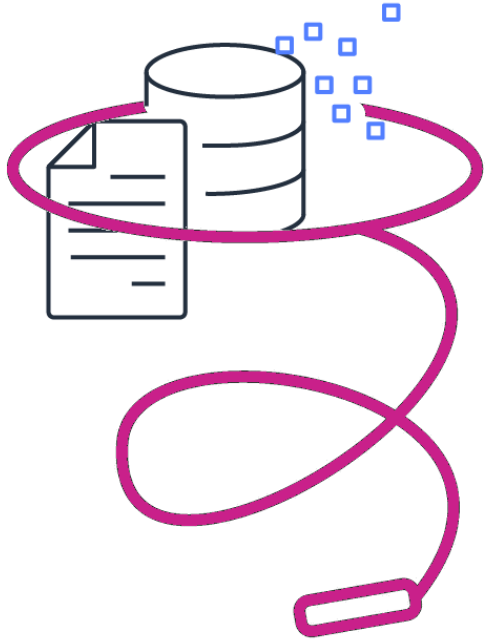


- Ingestion involves pulling data into the pipeline and applying transformations on the data.
- Ingestion by using an ETL flow works well with structured data that is destined for a data warehouse or other structured store.
- Ingestion by using an ELT flow works well for unstructured data that is destined for a data lake for a variety of use cases.
- ETL and ELT processing within a pipeline should evolve to optimize its value.

Data wrangling introduction

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Data wrangling



Transforming large amounts of unstructured or structured raw data from multiple sources with different schemas into a meaningful set of data that has value for downstream processes or users

Data wrangling addresses managing data variety



On-premises databases or file stores



Public datasets



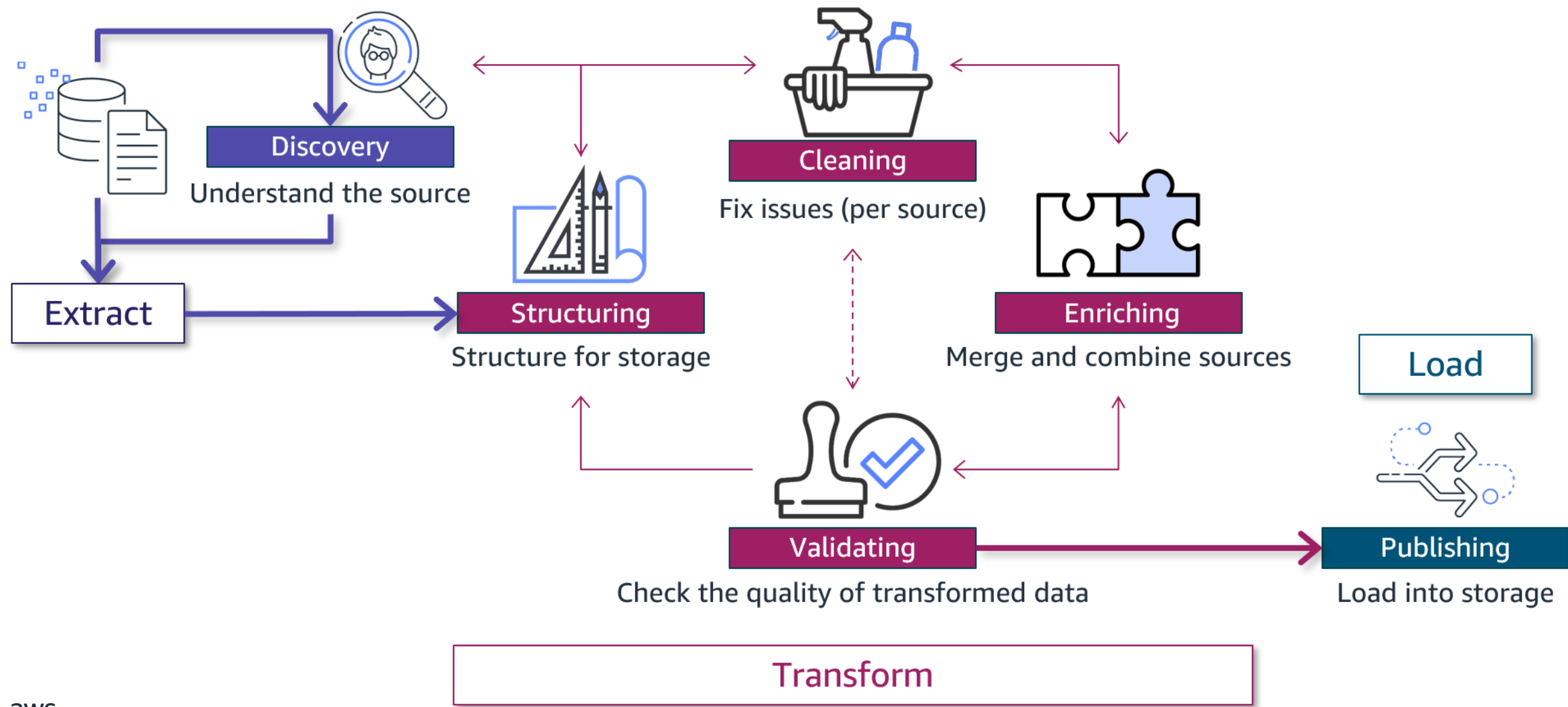
Events, Internet of Things (IoT) devices, sensors

A healthcare company has structured patient data.

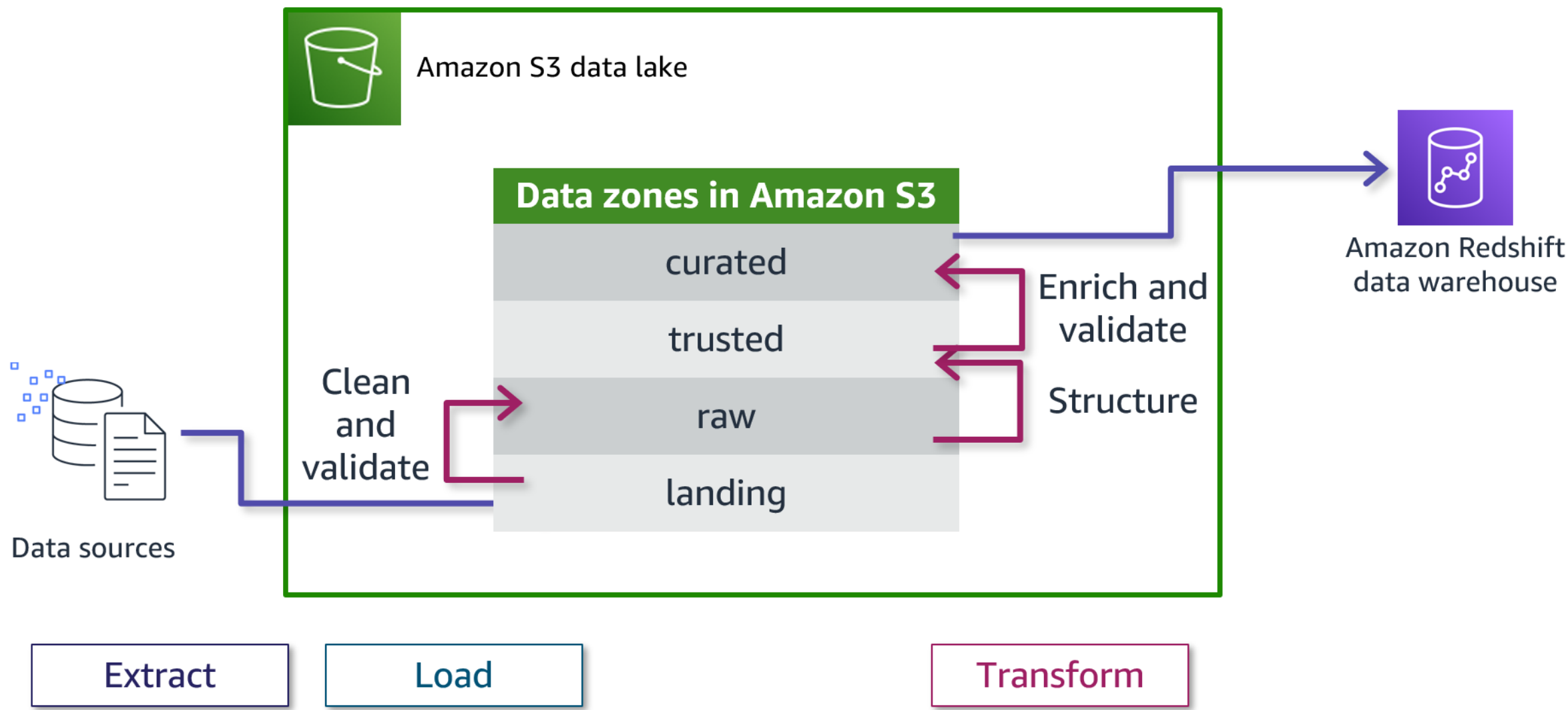
They want to combine patient data with semistructured data from a public dataset.

They also want to include unstructured data from a mobile app that logs real-time events.

Data wrangling steps



Example: ELT and data wrangling in a modern data architecture



Key takeaways: Data wrangling introduction

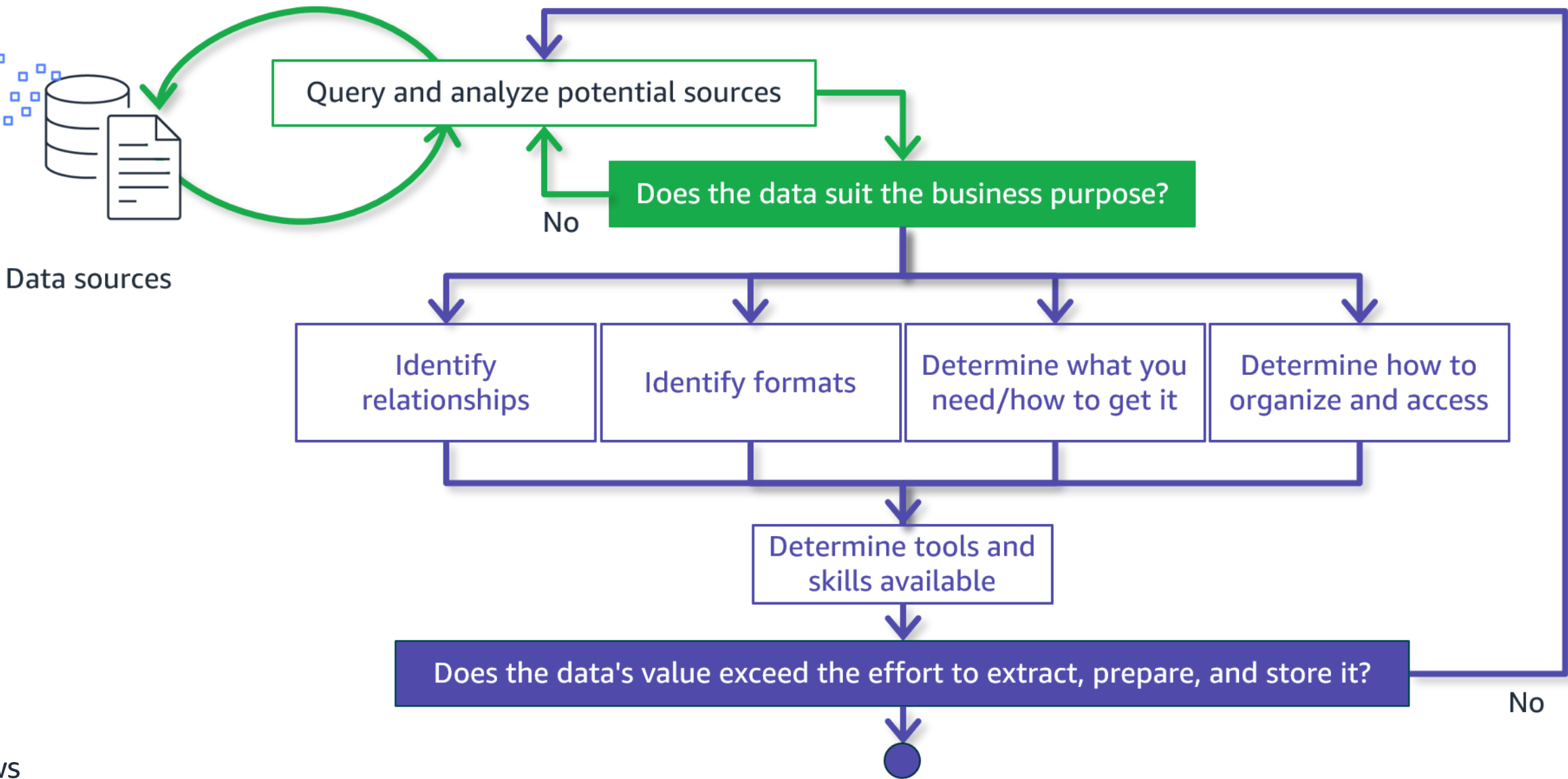


- Data wrangling describes the multi-step process of transforming large amounts of unstructured data or sets of structured data from multiple sources for an analytics use case.
- Data wrangling is especially important for data scientists when building ML models.
- Data wrangling steps might overlap, be iterative, or not occur at all in some ingestion processes.
- Data wrangling steps include discovery, structuring, cleaning, enriching, validating, and publishing.

Data discovery

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Decisions and tasks in discovery



Data discovery example scenario

- A company with a software as a service (SaaS) product recently acquired a startup company.
- The startup company's customer support team uses a different ticketing system than the company that acquired them.
- A data engineer has been asked by a data analyst to ingest support ticket data from the two products to perform an initial exploration of relationships between support experiences, ticket volumes, and contract renewals.
- The analyst wants to know the number and types of technical issues that have been worked on or closed for each customer in 2020. They also want each regional sales team to have access to only their customers.
- The existing sales analytics pipeline is built on AWS services and currently houses customer data in an Amazon Redshift data warehouse. However, it doesn't include customer support data.

Data discovery example: Query data sources

Ticket table in customer support system (supp1)

ticket_id	requestor_id	submitter	assignee	group	subject	status	priority	ticket_type	create_date	updated_date	solved_date
							High				
							Low				

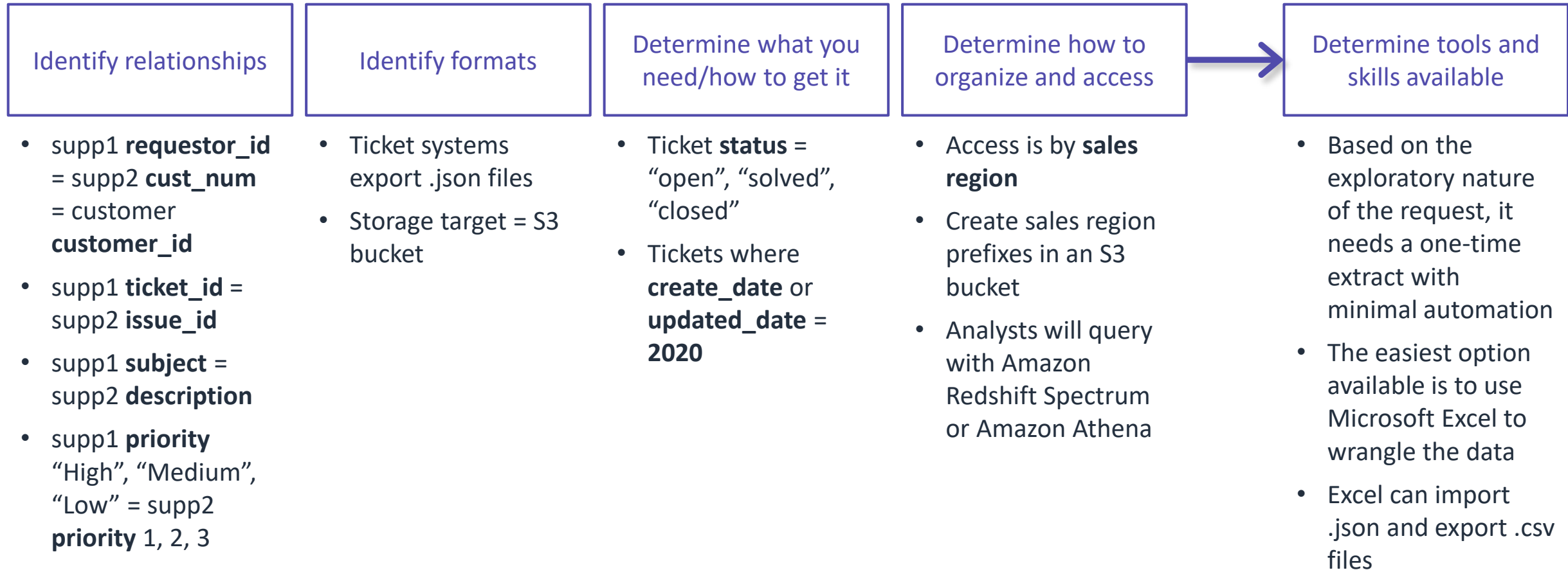
Ticket table in the acquired customer support system (supp2)

issue_id	cust_num	description	status	priority	create_date	updated_date	closed_date
				1			
				3			

Customer table in the data warehouse

customer_id	cust_name	primary_poc	status	sales_group

Data discovery tasks: Example scenario



Key takeaways: Data discovery

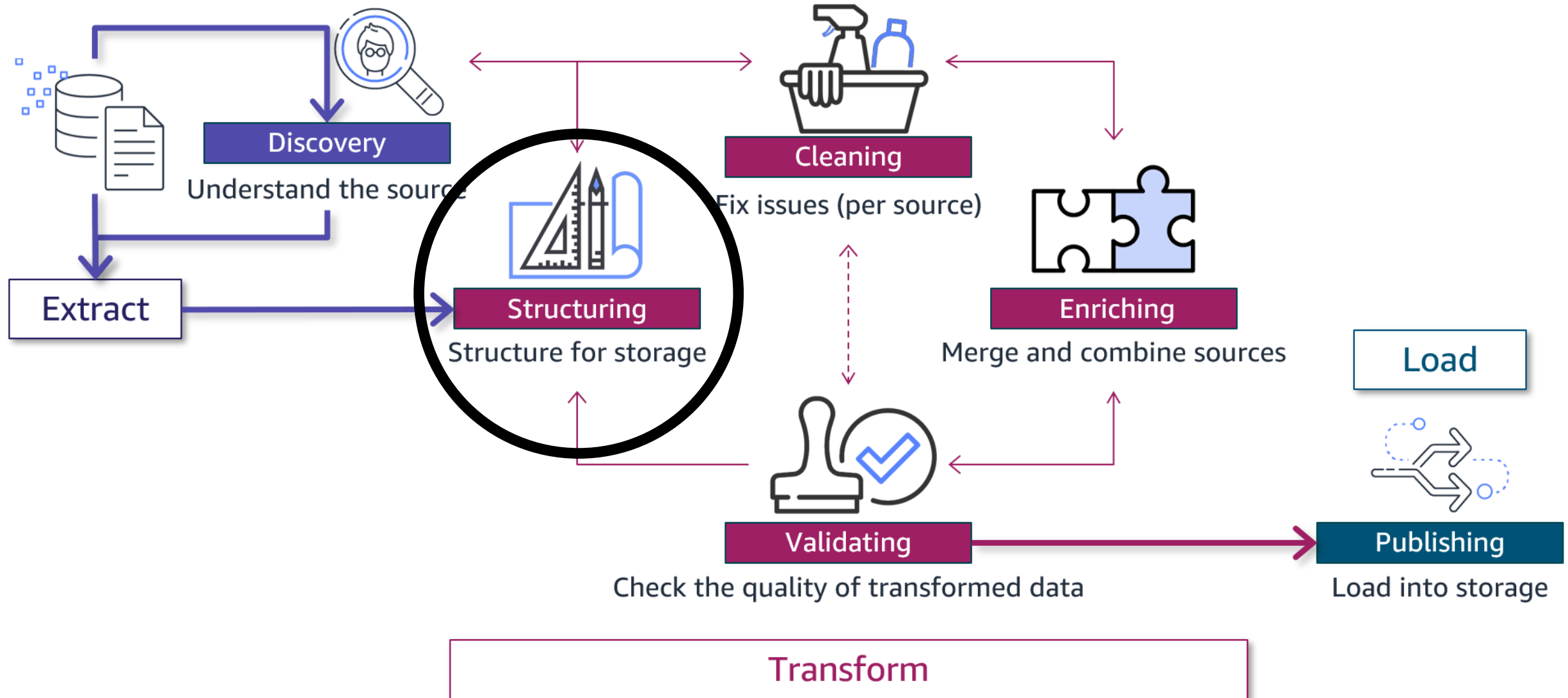


- Data discovery is iterative and might have a different focus depending on your role.
- Tasks performed during the data discovery step include the following:
 - Identifying relationships and formats
 - Determining how to filter the incoming data and how to organize it in your target storage
- The two primary outcomes of the data discovery phase are identification of the tools and resources required and a decision to move forward.

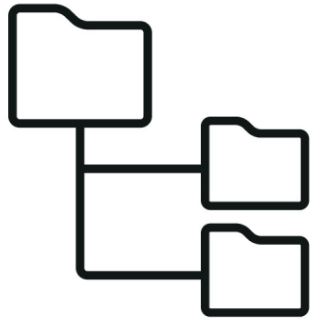
Data structuring

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The data structuring step within data wrangling



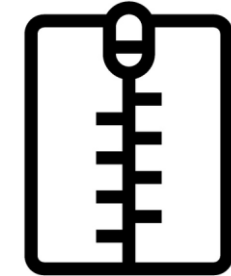
Decisions and tasks in data structuring



```
{  
  "ticket_id": 900864,  
  "requestor_id": 1800445,  
},
```

A line-art icon of a table with an arrow pointing to it from the JSON object. The table has a header row with 'SALES' and a second row with 'CUST'. Below these are several empty rows and columns, representing a data table.

SALES				
CUST				



Organize storage

- Control access.
- Create the folder structure.
- Establish partitions.

Parse the source file

- Convert strings and patterns into fields or attributes.

Map fields

- Match source fields to target fields.

Manage file size

- Split or merge files.
- Compress files.

Data structuring example: Parse supp2 file and map fields

Import .json and let Excel parse the file.

Update column headings to map to supp1 and data warehouse field names.

A	B	C	D	E	F	G	H
issue_id ticket_id	cust_num customer_id	description subject	status	priority	create_date	updated_date	closed_date solved_date
900865		Error message 808 <*&#	Closed	2	6/26/20 2:45:29	6/27/20 21:31:48	6/27/20 23:30:09
900866	2020411	Nullpointer exception on save	Open	1	4/2/20 12:15	4/5/20 11:15	
900867	2022010	Question about grades report	Open	1	4/6/20 10:56		

Key takeaways: Data structuring

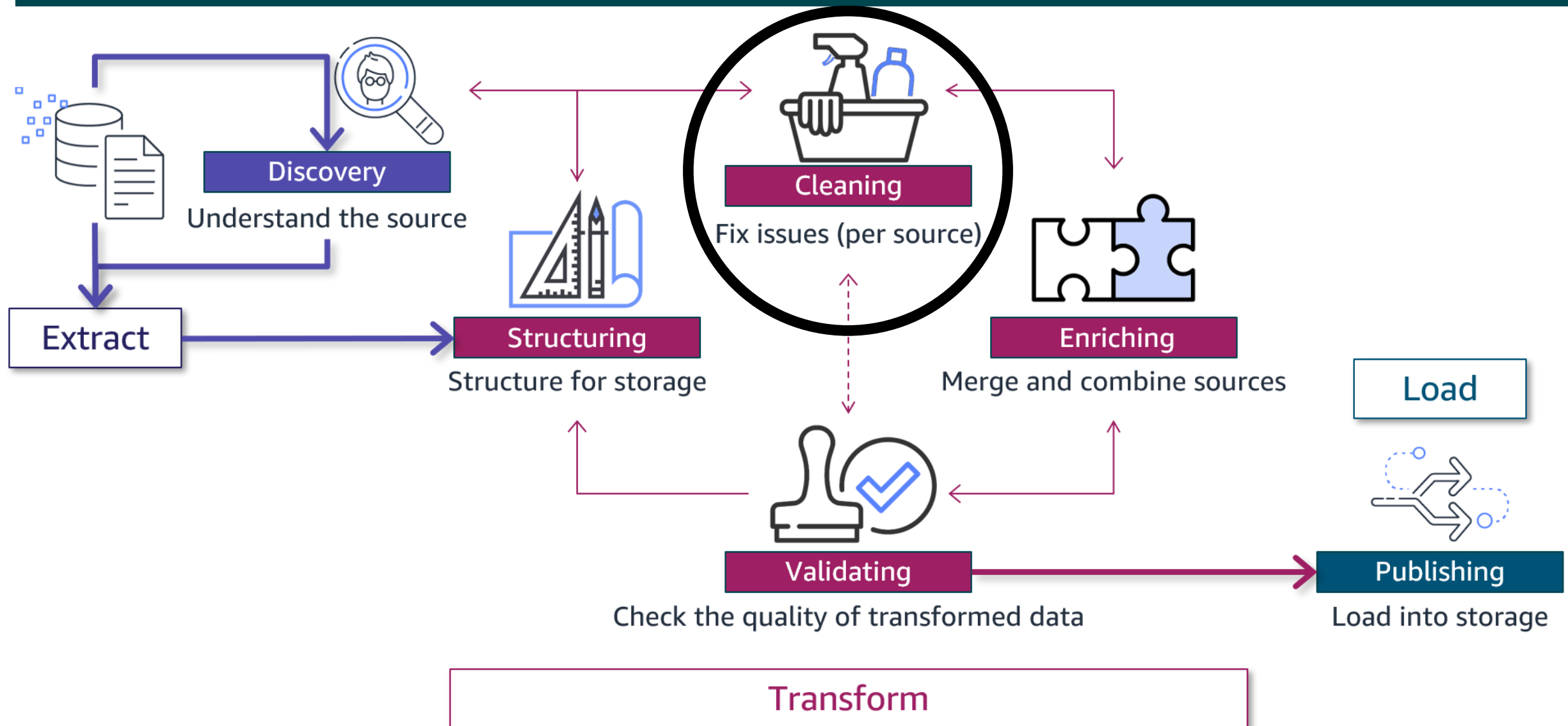


- Data structuring is about mapping raw data from the source file or files into a format that supports combining it and storing it with other data.
- Data structuring includes organizing storage and access, parsing the source file, and mapping the source file to the target.
- Data structuring also includes strategies and methods to optimize file size, such as splitting or compressing files.

Data cleaning

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The cleaning step within data wrangling



Decisions and tasks in cleaning



001	42	4/2	abc
002		4/5	abc
003	78	5/5	
004	14	5/8	def
005	4200	5/9	def

Remove data

- Remove unwanted columns.
- Remove duplicate values.
- Remove "garbage" values.

001	42	4/2
002	0	4/5
003	78	5/5
004	14	5/8
005	4200	5/9

Fill missing data

- Fill null columns.
- Fill missing values for mandatory fields.



001	42	22-04-02
002	0	22-04-05
003	78	22-05-05
004	14	22-05-08
005	4200	22-05-09

Check data types

- Validate or modify data types.

001	42	22-04-02
002	0	22-04-05
003	78	22-05-05
004	14	22-05-08
005	42	22-05-09

Fix outliers

- Identify outliers, and either remove or fix data as appropriate.

Data cleaning example: Remove and fix data

Replace blanks in the required **customer_id** field with a placeholder.

Remove invalid characters from the **subject** field.

Replace the numeric **priority** field with the text field equivalent of supp1.

Modify the timestamps to conform to the data warehouse date format.

A	B	C	D	E	F	G	H
ticket_id	customer_id	subject	status	priority	create_date	updated_date	solved_date
900865	9999999	Error message 808 ←*%&#	Closed	Medium	2020-06-26 02:45:29	2020-06-27 21:31:48	2020-06-27 23:30:09
900866	2020411	Nullpointer exception on save	Open	High	2020-04-02 12:15:00	2020-04-05 11:15:00	
900867	2022010	Question about grades report	Open	High	2020-04-06 10:56:00		

Key takeaways: Data cleaning

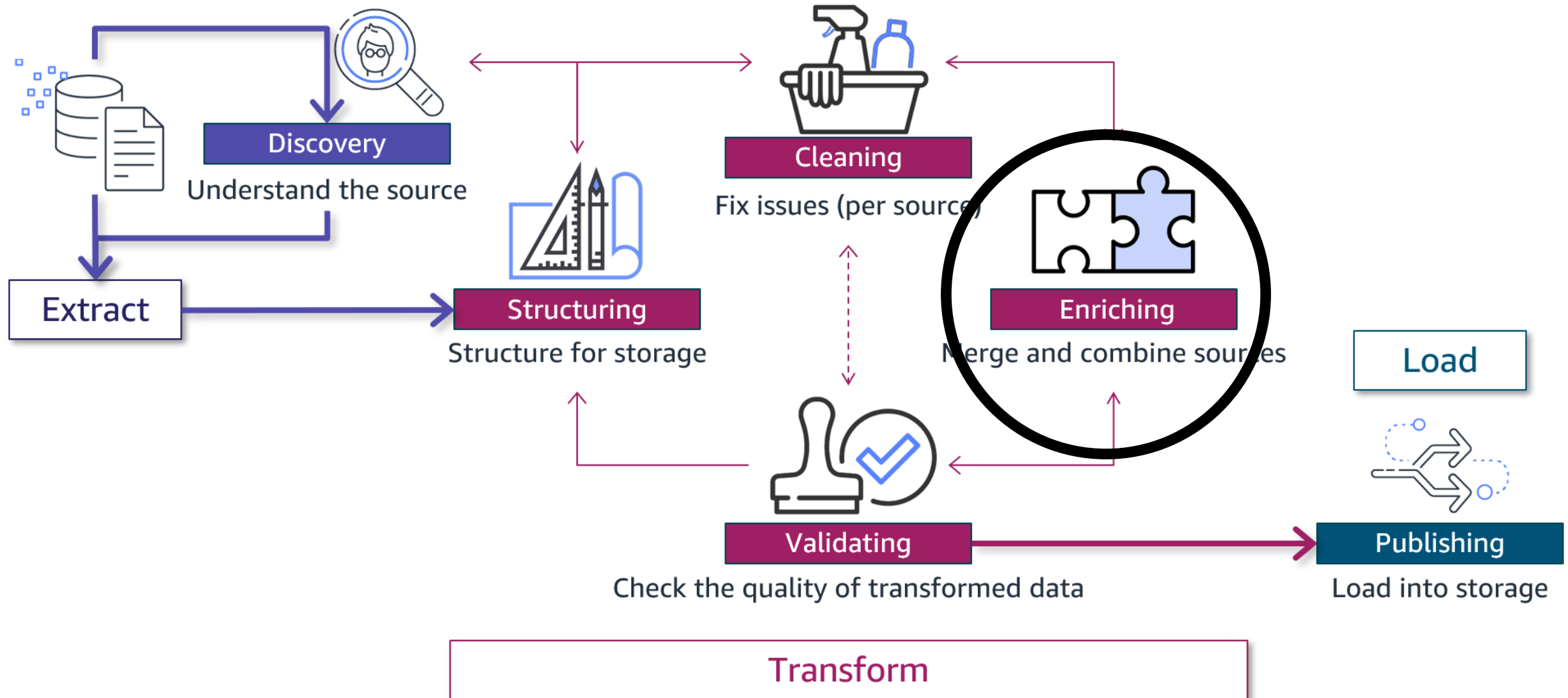


- Data cleaning is about preparing the source data for use in a pipeline.
- Cleaning is usually done for each source based on the characteristics of that source.
- Cleaning includes tasks such as removing unneeded, duplicate, or invalid data, and fixing missing values and data types.
- How the cleaning tasks resolve issues depends on the role of the person who cleans the data.

Data enriching

Ingesting and Preparing Data

The enriching step within data wrangling

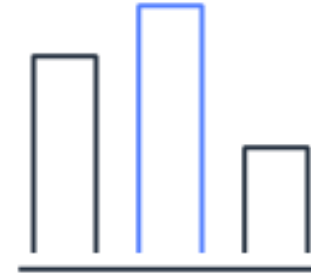


Decisions and tasks in enriching



Merge sources

- Combine data from cleaned source files into a single dataset.



Supplement data

- Add additional values to support analysis or desired visualization.

Data enriching example: Combine disparate sources

Combine rows from supp1 with rows from supp2.

Add a **ticket_type** category for supp2 rows.

Add the sales region.

A	B	C	D	E		F	G	H	
ticket_id	customer_id	subject	status	priority	ticket_type	create_date	updated_date	solved_date	sales_region
900865	9999999	Error message 808	Closed	Medium	Problem	2020-06-26 02:45:29	2020-06-27 21:31:48	2020-06-27 23:30:09	
900866	2020411	Nullpointer exception on save	Open	High	Problem	2020-04-02 12:15:00	2020-04-05 11:15:00		USEast
900867	2022010	Question about grades report	Open	High	Question	2020-04-06 10:56:00			USWest
900862	1744899	Nightly batch failing	Closed	Medium	Problem	2020-01-04 08:06:00	2020-04-31 09:00:00	2020-04-31 09:00:00	Europe
900865	1744005	Intermittent error message	Open	Low	Problem	2020-01-06 11:14:00			Europe

Key takeaways: Data enriching

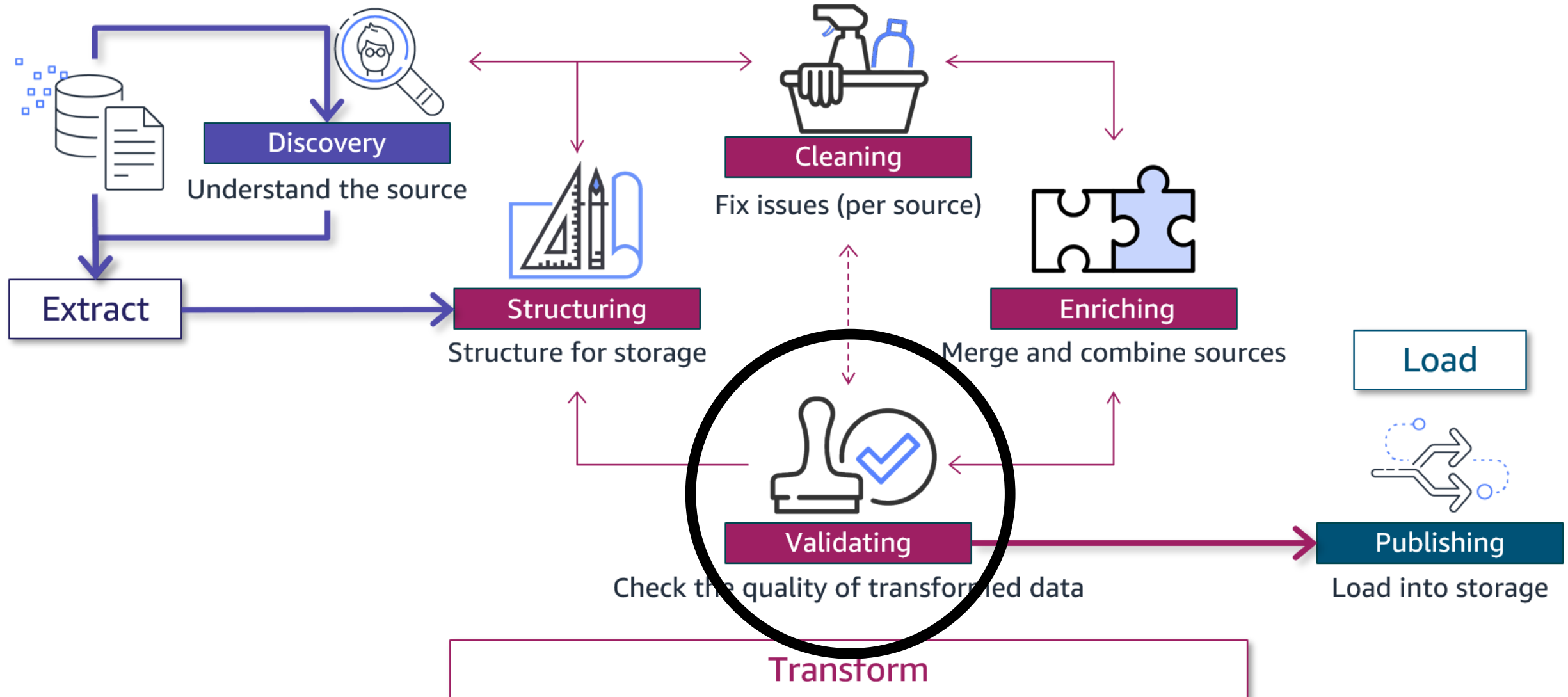


- Data enriching is about combining your data sources together and adding value to the data.
- During enriching, you might merge sources, add additional fields, or calculate new values.

Data validating

Ingesting and Preparing Data

The validating step within data wrangling



Decisions and tasks in validating



Audit your work

- Count expected rows.
- Check consistency.
- Check expected formats and data types.
- Check for duplicates.
- Check for PII.
- Check for outliers.



Fix the data

- Address audit findings.
- Address issues at the source.
- Fix issues with integration tools or scripts.

Data validating example

Address duplicate ticket IDs.

Find the correct **customer_id**, fix in the source, and run a script to get **sales_region**.

A	B	C	D	E		F	G	H	
ticket_id	customer_id	subject	status	priority	ticket_type	create_date	updated_date	solved_date	sales_region
900865	2070555	Error message 808	Closed	Medium	Problem	2020-06-26 02:45:29	2020-06-27 21:31:48	2020-06-27 23:30:09	USEast
900866	2020411	Nullpointer exception on save	Open	High	Problem	2020-04-02 12:15:00	2020-04-05 11:15:00		USEast
900867	2022010	Question about grades report	Open	High	Question	2020-04-06 10:56:00			USWest
900862	1744899	Nightly batch failing	Closed	Medium	Problem	2020-01-04 08:06:00	2020-04-31 09:00:00	2020-04-31 09:00:00	Europe
900865	1744005	Intermittent error message	Open	Low	Problem	2020-01-06 11:14:00			Europe

Count total rows and check against rows from individual systems.

Supp2 rows()=3
Supp1 rows()=2
Target rows()=5

Key takeaways: Data validating

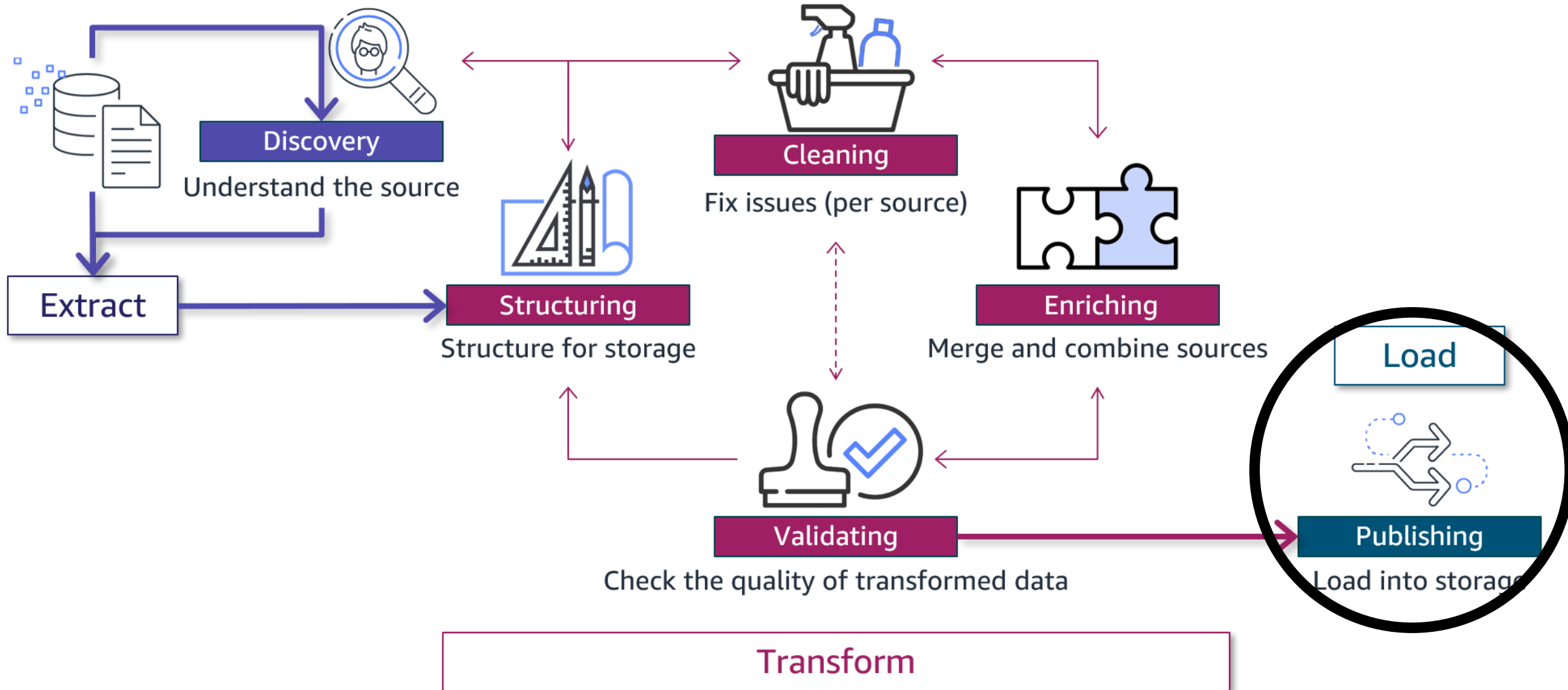


- Data validating is about ensuring the integrity of the dataset that you have created.
- Validating tasks might overlap with cleaning tasks.
- This step might be iterative as you check the results of your work, find issues, and address them.

Data publishing

Ingesting and Preparing Data

The publishing step within data wrangling



Decisions and tasks in publishing



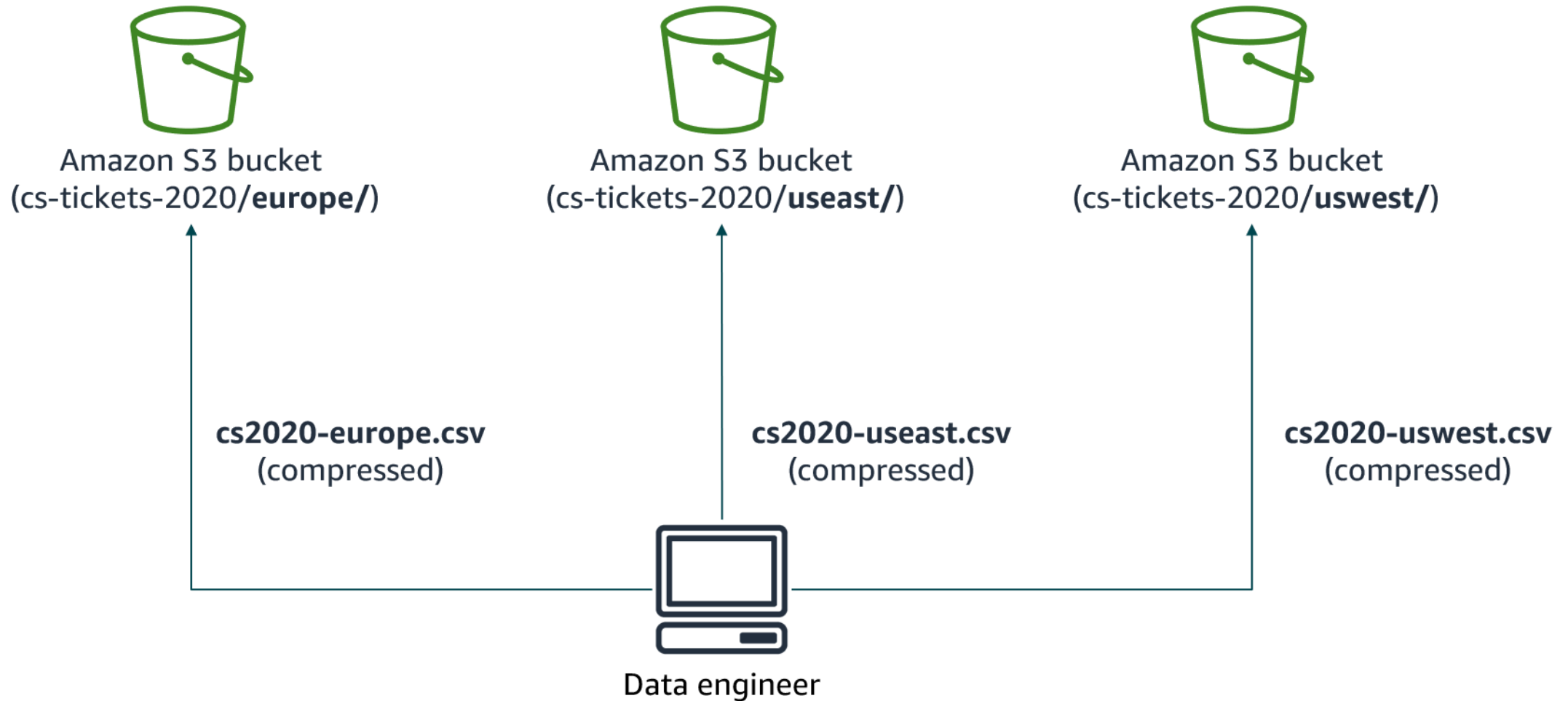
Move data into permanent storage

- Apply file management techniques (formats, compression, organization).
- Securely transfer data from source to destination.
- Write the dataset or datasets to the correct storage locations.

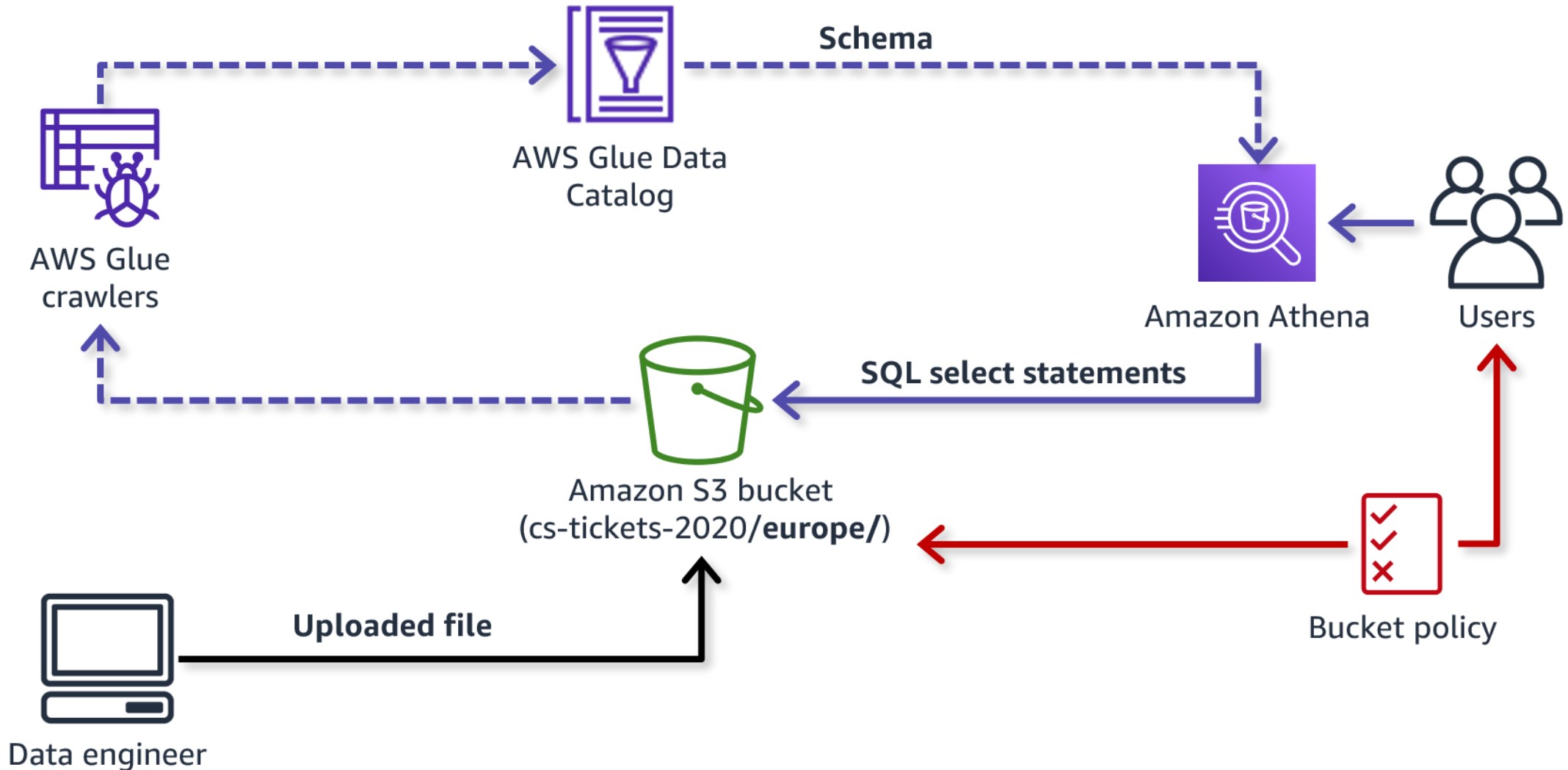
Make data available to consumers

- Apply appropriate access controls to data.
- Save metadata about the dataset.
- Configure how often and when to run the extract.
- Log the update history.

Data publication example: Moving data to permanent storage



Data publication example: Making the data available



Key takeaways: Data publishing



- Data publishing is about making the dataset available for use.
- In this step, you move the data to the storage layer of the pipeline. The decisions that you make during the structuring step about organization and file management are used to load the data into storage.
- At this step, you also need to provide methods to update the dataset and monitor ongoing ingestion.

Module wrap-up

Ingesting and Preparing Data

Module summary

This module prepared you to do the following:

- Distinguish between the processes of ETL and ELT.
- Define data wrangling in the context of ingesting data into a pipeline.
- Describe key tasks in each of these data wrangling steps:
 - Discovery
 - Structuring
 - Cleaning
 - Enriching
 - Validating
 - Publishing

Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

A data engineer has been asked to provide data to support a new sales report. The report will combine sales data for four different products, which have been tracked in four different systems. The resulting data needs to be segmented into regions, and each regional dataset should be available only to sales teams for the region.

What should the data engineer do first to meet this request?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- **Four different systems**
- Data **should be available only to sales teams for the region** where the sale occurred
- What should the data engineer do **first** to meet this request?

Sample exam question: Response choices

A data engineer has been asked to provide data to support a new sales report. The report will combine sales data for four different products, which have been tracked in **four different systems**. The resulting data needs to be segmented into regions, and each regional dataset **should be available only to sales teams for the region**.

What should the data engineer do **first** to meet this request?

Choice	Response
A	Create an S3 bucket with folders for each product.
B	Export data from each system, and merge the data into a single file.
C	Identify the relationships between fields across each of the sources.
D	Replace null values in each source system field with zeroes.

Sample exam question: Answer

The correct answer is C.

Choice	Response
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C	Identify the relationships between fields across each of the sources.
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Thank you



Corrections, feedback, or other questions?

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